

PREDICTION: LEGITIMATE

RISK SCORE: 54 / 100 (MEDIUM RISK)

Scam Probability: 1.04% | Model Confidence: 98.96%

Banking Action: CONTEXTUAL WARNING & 2FA CHALLENGE

Official Recommendation: Additional Verification Recommended. Some unusual characteristics were detected (elevated transfer amount or novel transfer window). Please review the recipient name, account number, and amount before submitting your biometric verification.

1. Transaction Details & Actual Parameter Values

|                  |                   |                          |       |
|------------------|-------------------|--------------------------|-------|
| Transaction ID:  | TXN-1789626084138 | Transaction Amount:      | 150.0 |
| Customer ID:     | C123456789        | Sender Old Balance:      | None  |
| Beneficiary ID:  | M987654321        | Sender New Balance:      | None  |
| Payment Type:    | PAYMENT           | Receiver Old Balance:    | None  |
| Device Type:     | None              | Receiver New Balance:    | None  |
| Location Region: | None              | 6h Transaction Velocity: | None  |
| New Beneficiary: | No                | Prior Fraud History:     | No    |

2. Composite Risk Score Breakdown & Latency Timing

| Risk Dimension             |  | Sub-Score | Status & Impact              |  |  |
|----------------------------|--|-----------|------------------------------|--|--|
| Model Fraud Probability    |  | 1.04%     | Weight: 70% (Score: 0.0)     |  |  |
| Amount Risk                |  | 24.0 pts  | Transaction size evaluation  |  |  |
| Velocity Risk              |  | 10.0 pts  | 6-hour transaction frequency |  |  |
| Device Risk                |  | 10.0 pts  | Device profile verification  |  |  |
| Location Risk              |  | 10.0 pts  | Geographic regional risk     |  |  |
| Beneficiary Risk           |  | 65.0 pts  | New beneficiary check        |  |  |
| Historical Behaviour       |  | 24.0 pts  | Prior fraud flag check       |  |  |
| Balance Behaviour          |  | 0.0 pts   | Account wipeout detection    |  |  |
| FINAL COMPOSITE RISK SCORE |  | 54 / 100  | MEDIUM RISK CATEGORY         |  |  |

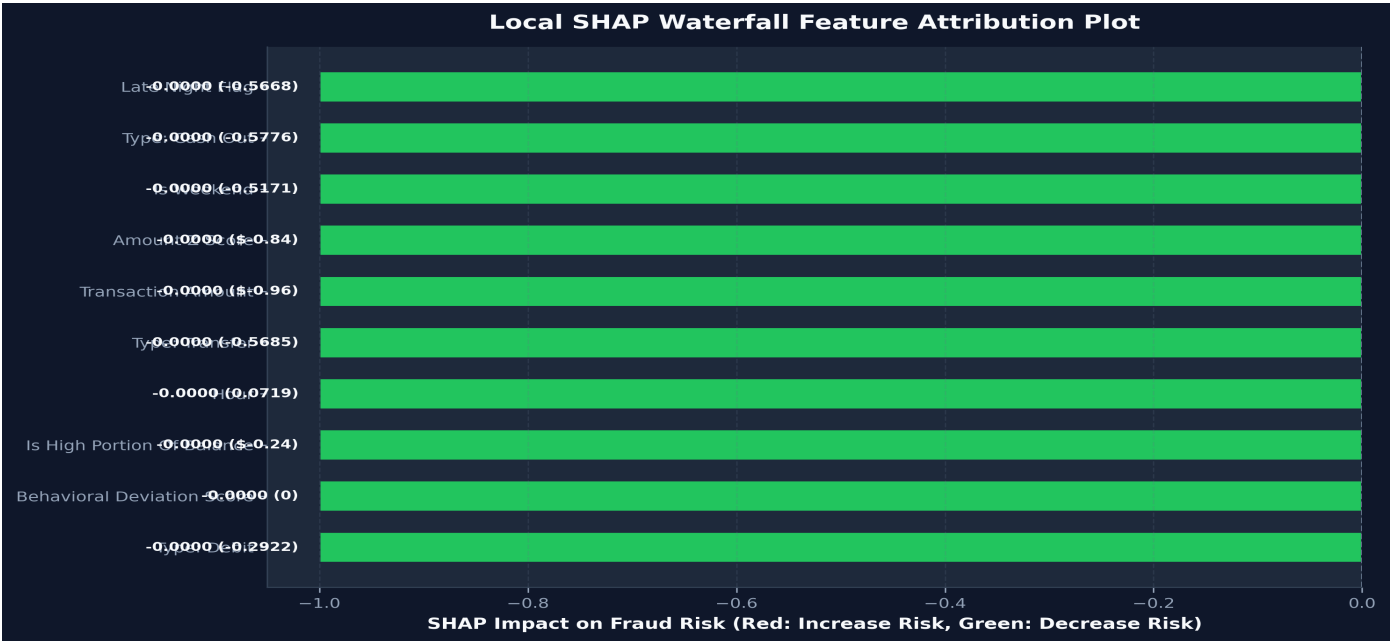
|                      |                      |                       |                        |                    |                   |
|----------------------|----------------------|-----------------------|------------------------|--------------------|-------------------|
| Feature Eng: 6.84 ms | Prediction: 12.31 ms | SHAP Engine: 28.99 ms | LIME Engine: 156.06 ms | Rendering: 0.00 ms | TOTAL: 1424.55 ms |
|----------------------|----------------------|-----------------------|------------------------|--------------------|-------------------|

3. Banking-Grade Analyst Narrative Report

|                       |                  |
|-----------------------|------------------|
| Transaction Summary:  |                  |
| Behaviour Analysis:   |                  |
| High Risk Indicators: | None identified. |
| Safe Indicators:      | None identified. |
| Model Explanation:    |                  |
| SHAP Summary:         |                  |
| Final Recommendation: |                  |

4. Top 10 SHAP Feature Contribution Attributions

| Feature Name               | Actual Value | SHAP Value | Contrib % | Direction      | Impact Level |
|----------------------------|--------------|------------|-----------|----------------|--------------|
| Transaction Amount         | \$-0.96      | -0.0000    | 3.76%     | Decreases Risk | Low          |
| Is Weekend                 | -0.5171      | -0.0000    | 3.76%     | Decreases Risk | Low          |
| Late Night Flag            | -0.5668      | -0.0000    | 3.76%     | Decreases Risk | Low          |
| Type: Cash Out             | -0.5776      | -0.0000    | 3.76%     | Decreases Risk | Low          |
| Type: Transfer             | -0.5685      | -0.0000    | 3.76%     | Decreases Risk | Low          |
| Amount Z Score             | \$-0.84      | -0.0000    | 3.76%     | Decreases Risk | Low          |
| Is High Portion Of Balance | \$-0.24      | -0.0000    | 3.76%     | Decreases Risk | Low          |
| Hour                       | 0.0719       | -0.0000    | 3.76%     | Decreases Risk | Low          |
| Behavioral Deviation Score | 0            | -0.0000    | 3.76%     | Decreases Risk | Low          |
| Amount To Oldbalance Ratio | \$-0.31      | -0.0000    | 3.76%     | Decreases Risk | Low          |



5. Local LIME Decision Rules & Explanations

| Decision Rule Condition | Rule Weight | Effect Direction | Domain Explanation |
|-------------------------|-------------|------------------|--------------------|
|-------------------------|-------------|------------------|--------------------|

|                                              |         |                |                                                                                                        |
|----------------------------------------------|---------|----------------|--------------------------------------------------------------------------------------------------------|
| Account Balance Wipeout = Yes (100% Drained) | +0.2800 | Increases Risk | Sender account balance drained to zero or near-zero post-transfer.                                     |
| amount <= -0.96                              | +0.0000 | Decreases Risk | Transaction amount is within low-risk customer spending bounds.                                        |
| log_amount <= -1.84                          | +0.0000 | Decreases Risk | Transaction amount is within low-risk customer spending bounds.                                        |
| oldbalanceOrg <= -1.43                       | +0.0000 | Decreases Risk | Sender account maintains sufficient baseline capital.                                                  |
| log_oldbalanceOrg <= -1.56                   | +0.0000 | Decreases Risk | Sender account maintains sufficient baseline capital.                                                  |
| amount_to_oldbalance_ratio <= -0.31          | +0.0000 | Decreases Risk | Transaction amount is within low-risk customer spending bounds.                                        |
| is_high_portion_of_balance <= -0.24          | +0.0000 | Decreases Risk | Feature condition 'is_high_portion_of_balance <= -0.24' supported legitimate transaction verification. |
| hour <= 0.07                                 | +0.0000 | Decreases Risk | Feature condition 'hour <= 0.07' supported legitimate transaction verification.                        |
| day_of_week <= -1.43                         | +0.0000 | Decreases Risk | Feature condition 'day_of_week <= -1.43' supported legitimate transaction verification.                |
| is_weekend <= -0.52                          | +0.0000 | Decreases Risk | Feature condition 'is_weekend <= -0.52' supported legitimate transaction verification.                 |

